

## I-DECTIN 0.5% Pour on

I-Dectin 0.5% Pour On appears to be a deep blue clear liquid.

### Composition

Each ml contains Ivermectin 5 mg

### Pharmacodynamics

Ivermectin is a mixture of two compounds belonging to the avermectin family, which are a macrocyclic lactone group of endectocides. Avermectin is a microbial metabolite of the soil organism *Streptomyces avermilitis*.

It is generally accepted that ivermectin exerts its action in two main ways, interference with neurotransmission and opening chloride ion channels. The effect of ivermectin on the parasitic CNS is considered to operate through glutamate-mediated chloride channels. Compounds of this class may also interact with other ligand-gated chloride channels, such as those gated by the neurotransmitter gamma-aminobutyric acid (GABA).

The opening of pre-synaptic chloride ion channels results in an efflux of chloride ions and depolarisation of the nerve terminal. These effects interfere with normal neurotransmission between nerves and muscles, resulting in parasite paralysis and eventual death.

### Pharmacokinetics

After administration of the product, the ivermectin is absorbed through the skin into the circulation of the treated animal. The maximum concentration in plasma occurs around 70 hours after application. Peak concentrations of about 7 ng/ml are obtained.

The residual antiparasitic effect of ivermectin is due to its persistence, which in turn is due in part to its long intrinsic half-life ( $t_{1/2\beta}$  of approximately 210 hours), in part to its relatively high plasma protein binding (80% in cattle; binding remains relatively constant over time) and in part to the nature and type of the ivermectin formulation.

Elimination is in the faeces (via biliary excretion). Over 60% of the dose is excreted after 3 days.

### Indications

For the treatment of the following species of gastrointestinal roundworms, lungworms, warbles, mites and lice:

#### Gastro-intestinal roundworms (adults and fourth stage larvae):

*Ostertagia ostertagi* (including inhibited O ostertagi)

*Haemonchus placei*

*Trichostrongylus axei*

*Trichostrongylus colubriformis*

*Cooperia* spp

*Oesophagostomum radiatum*

*Strongyloides papillosus* (adult)

#### Lungworms (adult and fourth stage larvae):

*Dictyocaulus viviparus*

#### Eye worms (adult):

*Thelazia* spp

#### Warbles (parasitic stages):

*Hypoderma bovis*

*Hypoderma lineatum*

#### Mites:

*Chorioptes bovis* (reduction of infestation)

*Sarcoptes scabiei* var *bovis*.

#### Sucking lice:

*Linognathus vituli*,

### Warning and Precautions

Care should be taken to avoid the following practices because they increase the risk of development of resistance and could ultimately result in ineffective therapy:

-Too frequent and repeated use of anthelmintics from the same class, over an extended period of time.

-Underdosing, which may be due to underestimation of body weight, misadministration of the product, or lack of calibration of the dosing device.

Suspected clinical cases of resistance to anthelmintics should be further investigated using appropriate tests (e.g. Faecal Egg Count Reduction Test). Where the results of the test(s) strongly suggest resistance to a particular anthelmintic, an anthelmintic belonging to another pharmacological class and having a different mode of action should be used.

#### Special precautions for use

##### *Special precautions for use in animals*

Ivermectins may not be well tolerated in all non-target species. Cases of intolerance with fatal outcome are reported in dogs, especially Collies, Old English Sheepdogs and related breeds or crosses, and also in turtles/tortoises. To avoid secondary reactions due to the death of Hypoderma larvae in the oesophagus or in the spine, it is recommended to administer the product at the end of the period of warble fly activity and before the larvae reach their resting site. As Ivermectin is extremely dangerous to fish and aquatic life, treated animals should not have direct access to surface water and ditches during treatment.

##### *Special precautions to be taken by the person administering the veterinary medicinal product to animals*

The product may be irritating to human skin and eyes and the user should be careful not to apply it to himself or other persons. Operators should wear rubber gloves and boots with a waterproof coat when applying the product. Protective clothing should be washed after use. Use only in well-ventilated areas or outdoors.

HIGHLY FLAMMABLE.

Keep away from heat, spark, open flame or other source of ignition. Do not eat, drink or smoke whilst handling the product.

As absorption through skin can occur, in the event of accidental skin contact the affected area should be washed immediately with soap and water.

If accidental eye exposure occurs, flush the eyes immediately with water and seek medical attention. Wash hands after use.

#### Statement on Usage During Pregnancy and Lactation

The product can be used during pregnancy and lactation. The product will not affect the fertility of cows and bulls and can be given to all ages of animals including young calves.

#### Interactions with Other Medicaments

Do not combine ivermectin treatment with vaccination against lungworms. If vaccinated animals are to be treated, treatment should not be carried out within a period of 28 days before or after vaccination. This veterinary medicinal product must not be mixed with other products.

#### Adverse Effects

Occasionally slight irritation at the application site may occur. However, usually these irritations rapidly disappear without treatment.

#### Overdose and Treatment

No sign of toxicity appeared up to 1.5 mg/kg (3 times the recommended dose rate). No antidote has been identified. The signs of overdose can

*Haematopinus eurysternus*

**Biting Lice:**

*Damalinia bovis*

The product given at the recommended dosage of 500 micrograms/kg bodyweight, has persistent activity against *Trichostrongylus axei* and *Cooperia* spp acquired during the 14 days after treatment, only if the whole herd is treated simultaneously. It also has a persistent activity against *Ostertagia ostertagi* and *Oesophagostomum radiatum* acquired during the first 21 days after treatment and *Dictyocaulus viviparus* (lungworm) acquired during the first 28 days after treatment. It also has a persistent activity on horn flies (*Haematobia irritans*) for 28 days after treatment, partial efficacy may last for up to 35 days post application. Occasionally variable activity may be observed against *Haemonchus placei* (L4), *Cooperia* spp, *Trichostrongylus axei* and *Trichostrongylus colubriformis*.

**Dosage and administration**

For single topical administration

Dosage: 1 ml per 10 kg bodyweight (based on a recommended dose of 500 micrograms/kg bodyweight).

To ensure administration of a correct dose, body weight should be determined as accurately as possible.

If animals are to be treated collectively rather than individually they should be grouped according to their bodyweight and dosed accordingly, in order to avoid under- or over- dosing.

Administration: Pour-on use. The formulation should be applied along the mid-line of the back in a narrow strip between the withers and tailhead. The interval between 2 treatments should be at least 28 days.

**Contraindication**

Do not use in cases of known hypersensitivity to the active ingredient.

The product has been formulated for topical application specifically for cattle.

It should not be administered to other species as severe adverse reactions, including fatalities in dogs, may occur.

be trembling, convulsions and coma. In case of overdose symptomatic treatment should be given.

**Withdrawal Period (s)**

Meat and offal: 28 days

Milk: Not permitted for use in lactating cattle producing milk for human consumption. Do not use in non-lactating dairy cows including pregnant dairy heifers within 60 days prior to calving.

**Storage Conditions**

Store below 30°C. Protect from sunlight and heat exposure.

Keep out of reach of children. (Jauhkan daripada kanak-kanak)

Any unused product or packaging derived from such veterinary medicinal products should be disposed of in accordance with local requirements.

Shelf life after first opening: 28 days.

Shelf life as packaged for sale: 3 years.

**Packaging Available**

500 ml.

**Product holder/ Manufacturer**

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